

Prince_AI: Dynamic Web Lab Generation for Core CSS Concepts

Initial Box Model Lab

Full prompt text

Act as a frontend web developer. Using your Canvas tool, Generate an interactive website that can be used for understanding the CSS Box Model and its relationship with the display property.

The page must have:

1. Two div elements, 'Box 1' and 'Box 2', so I can see how they interact. 'Box 1' will be the one we control.
2. The CSS must use different background colors for the content area, the padding area, and the margin area of 'Box 1' (e.g., using background-clip: content-box). The border should be a solid line.
3. A control panel with:
 - Sliders to control the padding, margin, border-width, and width of 'Box 1'.
 - Labels next to the sliders that show the current pixel value.
 - A Dropdown (select) to change the display property of 'Box 1' to: block, inline-block, and inline.
4. JavaScript that listens to all sliders and the dropdown, and updates the CSS properties of 'Box 1' in real-time.

Execution screenshot

CSS Box Model & Display Explorer

Code Preview Download

Diagnostic Mode Preview Mode

LEARNING PRESETS

Standard Block Pure Inline Text

Inline-Block Card Box-Sizing Comparison

PRIMARY CSS PROPERTIES

displayblock

box-sizingcontent-box

content-box (Standard default)

BOX METRICS SLIDERS

width180px

Legend: Margin Border Padding Content

PARENT CONTAINER (<div class="parent">)

Flow Direction: Left-to-Right

Sibling text before target.

Box 1 (Target)

Box 2 (Sibling)

Sibling text following boxes in normal page layout flow context.

Ask Gemini

CSS Box Model & Display Explorer

Code Preview Download

Diagnostic Mode Preview Mode

BOX METRICS SLIDERS

width180px

height100px

padding20px

border-width8px

margin24px

BOX 1 TEXT CONTENT

Box 1 (Target)

Box 2 (Sibling)

Sibling text following boxes in normal page layout flow context.

COMPUTED GEOMETRY

content-box

Content Box (W x H): 180px x 100px

Inner Border Box (W x H): 236px x 156px

Outer Total Area (W x H): 284px x 204px

Total Horizontal Space.

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LAYOUT ENGINE INSIGHT

BLOCK DISPLAY

Block elements break onto a new line and expand horizontally to fill 100% of their container width by default. They strictly respect all width, height, padding, and margin properties.

Notice how Box 1 pushes Box 2 down to the next line because block elements create vertical line breaks.

GENERATED CSS RULE

Copy

.box-1 {
 display: block;
 box-sizing: content-box;
 width: 180px;
 height: 100px;
 padding: 20px;
 border: 8px solid #eab308;
 margin: 24px;
 background-color: #3b82f6;
}

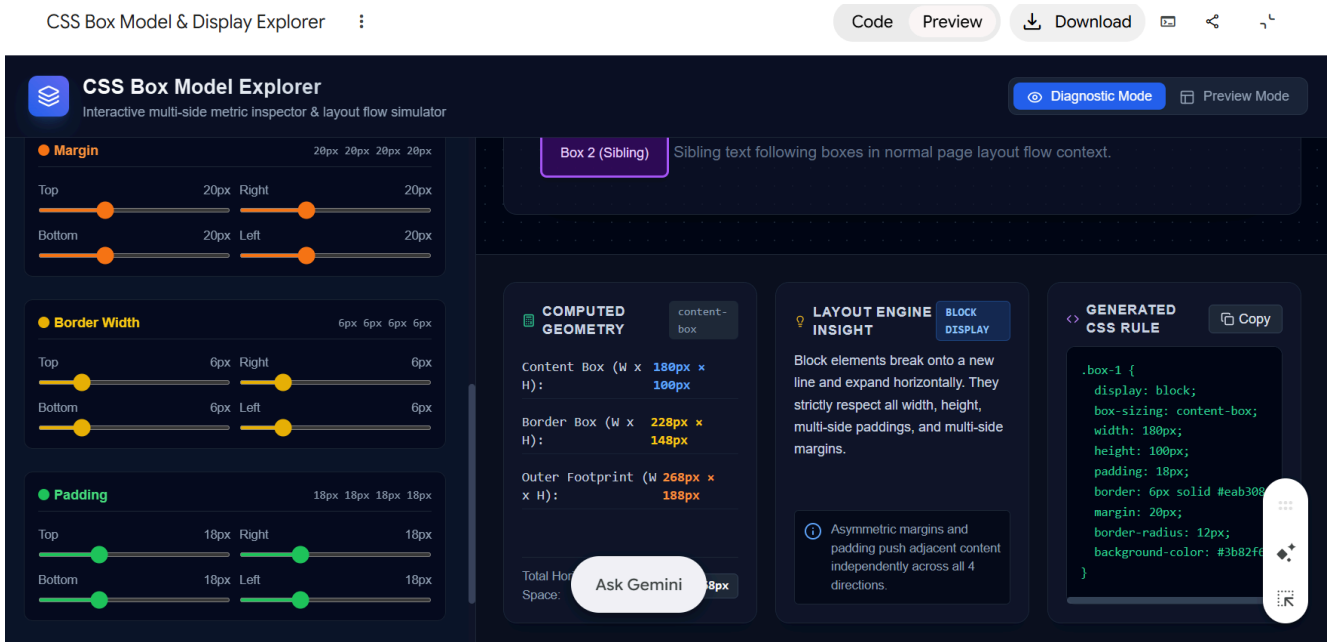
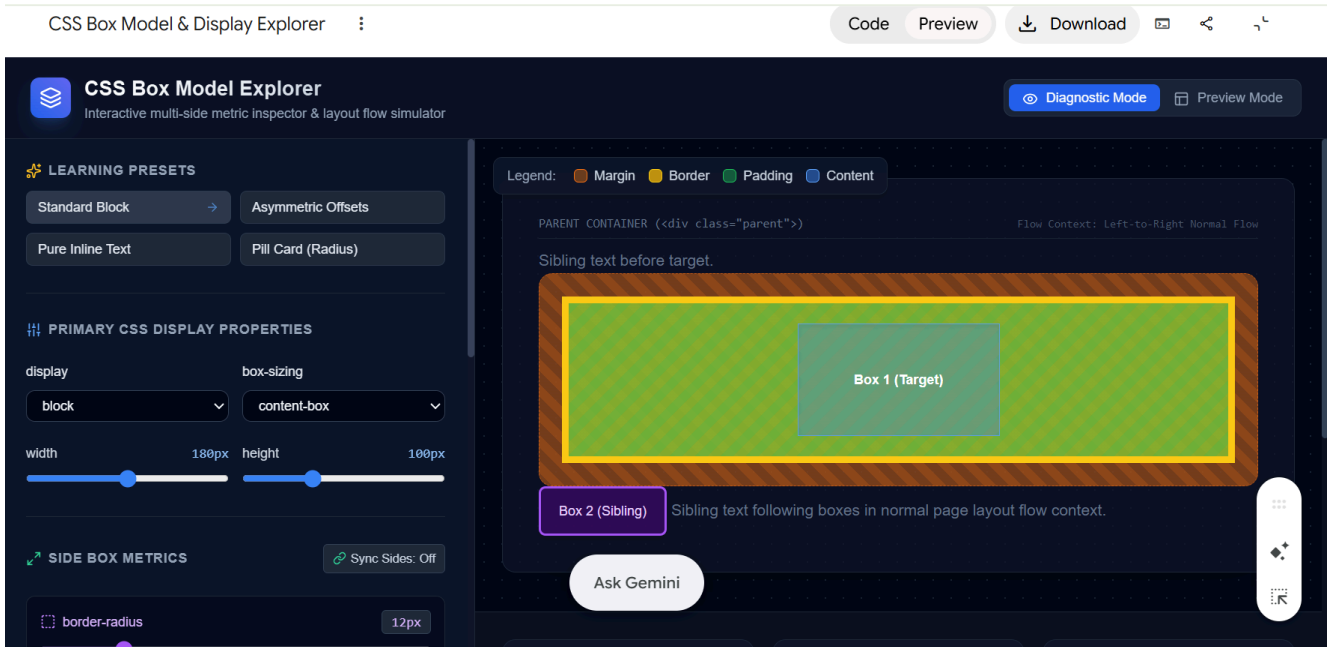
Refined Box Model Lab

Full prompt text

Implement sliders to adjust the margin, padding, and border for each side (top, right, bottom, left) individually, and add a separate slider for the corner radius.

Use this refinement as a follow-up with the initial Box Model source available as context. The refined file preserves the first version’s width and display controls and adds separate side controls and corner radius.

Execution screenshot



Flexbox and Grid Playground

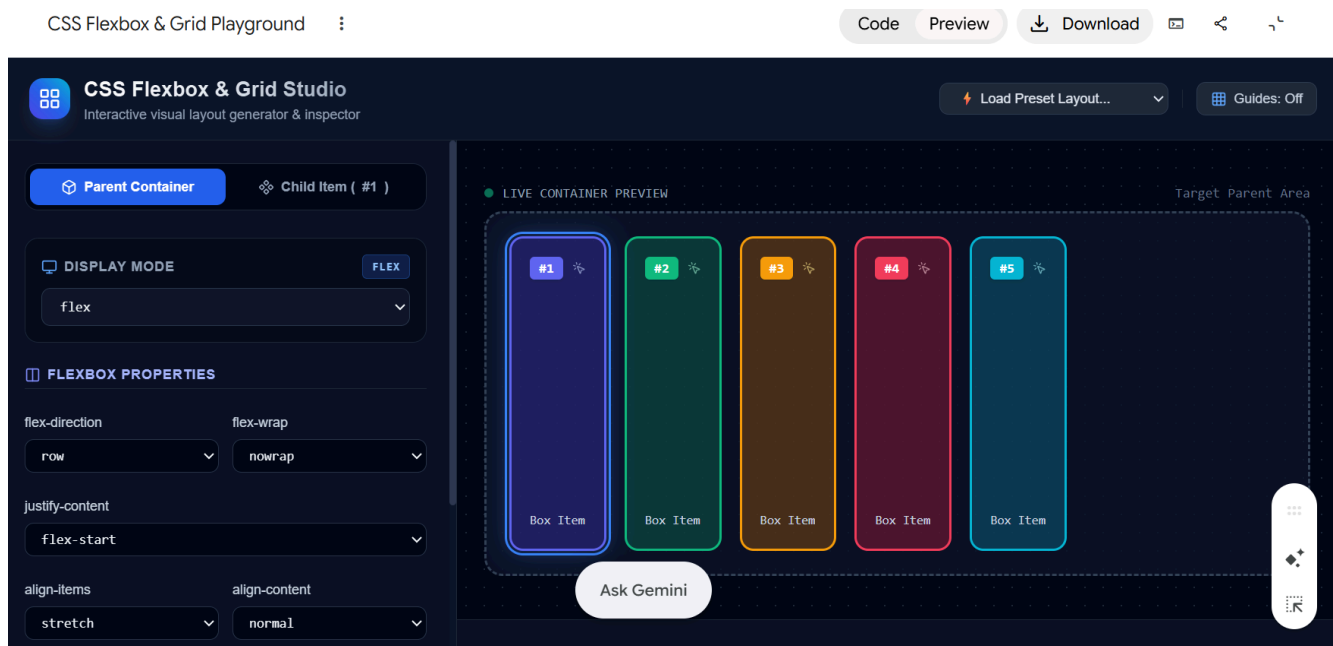
Full prompt text

Act as a frontend web developer. Using your Canvas tool, generate an interactive website that can be used as a playground for CSS Flexbox and Grid.

The page should have:

1. A `div` element acting as the container.
2. Several `div` elements inside acting as the items (e.g., 5 items).
3. Dropdown menus (selects) that allow me to change the CSS properties of the container.
4. I need to be able to change:
 - display (to switch between block, flex, and grid)
 - flex-direction (row, column)
 - justify-content (flex-start, center, space-between, etc.)
 - align-items (flex-start, center, stretch, etc.)
 - grid-template-columns (e.g., 1fr 1fr, 1fr 1fr 1fr)
5. The JavaScript must update the container's CSS in real-time when I change a dropdown.

Execution screenshot





CSS Flexbox & Grid Studio

Interactive visual layout generator & inspector

⚡ Load Preset Layout...

📱 Guides: Off

FLEXBOX PROPERTIES

flex-direction

row

flex-wrap

nowrap

justify-content

flex-start

align-items

stretch

align-content

normal

gap

16px

container padding

20px

+ Add Child Item

- Remove Item



GENERATED CSS RULE SET

📋 Copy CSS

```
.container {
  display: flex;
  gap: 16px;
  padding: 20px;
  flex-direction: row;
  flex-wrap: nowrap;
  justify-content: flex-start;
  align-items: stretch;
}
```

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Reflection and Synthesis

Draft reflection to review after trying the labs. Adjust the wording to match your own observations before submission.

Interactive labs make the relationship between CSS rules and browser behavior easier to examine than a static diagram alone. A diagram can identify the content, padding, border and margin, but a live control allows a learner to change one value and compare the result immediately. With content-box sizing, increasing padding or border width increases the outer size without changing the declared content width. Distinct colors help separate these areas, although the orange margin region must be treated as a visual guide because CSS margins themselves are transparent. Documentation remains useful for explaining the rules behind what the browser displays.

The display control is especially useful for understanding exceptions to the usual box model. For a non-replaced inline element such as this div, a declared width does not determine its rendered width. Left and right margins affect horizontal spacing, while top and bottom margins do not move surrounding lines. Vertical padding and borders still paint around inline content but do not increase the line-box height. Switching to inline-block makes width and all four margins effective while allowing neighboring boxes to share a line when space permits. In the separate playground, changing flex-direction also demonstrates that justify-content follows the main axis, while Grid provides explicit column tracks. These comparisons make the consequences of a CSS declaration observable.

Sequential prompting keeps the development process manageable. The first prompt establishes a working baseline with two boxes, general controls and live updates. The refinement prompt then changes a limited part of that baseline by introducing independent side controls and corner radius. Preserving both versions makes it possible to compare behavior and identify regressions, including accidentally losing an existing control. This resembles real software development, where a small implementation is reviewed, tested and extended as requirements become more specific. A single detailed prompt can be appropriate for a well-defined task, but smaller iterations make omissions easier to detect and corrections easier to isolate. Generated code still needs browser testing; a convincing explanation is not proof that every control works.

GitHub Repository Update

Version 1:

https://github.com/Prince-Ubakeeze/AirBnB_clone/blob/main/AI_Dynamic_Web_Lab/css_flexbox_grid_playground.html

Version 2:

https://github.com/Prince-Ubakeeze/AirBnB_clone/blob/main/AI_Dynamic_Web_Lab/css_flexbox_grid_playground%20v2.html

Version 3:

https://github.com/Prince-Ubakeeze/AirBnB_clone/blob/main/AI_Dynamic_Web_Lab/css_flexbox_grid_playground%20v3.html